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Section:- "C"

Ex:- "6.1"

* LAPLACE TRANSFORMATION:-

* Example "1":-

(1) let $f(t) = 1$ when $t \geq 0$ Find $F(s)$.

* Solution:-

From (1) we obtain by integration

$$\mathcal{L}(f) = \mathcal{L}(1) = \int_0^\infty e^{-st}\,dt = -\frac{1}{s}e^{-st}\Big|_0^\infty = \frac{1}{s}$$

Such an integral is called an "improper integral" and, by definition, is evaluated according to the rule.

$$\int_0^\infty e^{-st} f(t)\,dt = \lim_{T \to \infty} \int_0^T e^{-st} f(t)\,dt.$$

Hence our convenient notation meaning

$$\int_0^\infty e^{-st}\,dt = \lim_{T \to \infty} \left[\frac{-1}{s} e^{-st} \right]_0^T$$

$$\lim_{T \to \infty} \left[\frac{-1}{S} e^{-ST} + \frac{1}{S} e^{0} \right] = \frac{1}{S}$$

We shall use this notation throughout this chapter.

***Example "2":-**

① Let $f(t) = e^{at}$ when $t \geq 0$ where a is a constant. Find $\mathcal{L}(f)$.

***Solution:-**

$$\mathcal{L}(e^{at}) = \int_{0}^{\infty} e^{-st} e^{at} \, dt =$$

$$\mathcal{L}(e^{at}) = \frac{1}{a-S} e^{-(S-a)t} \Big|_{0}^{\infty}$$

$$\mathcal{L}(e^{at}) = \frac{1}{S-a}$$

* Example "3":-

Q) Find the transform of cosh at and Sinh at.

* Solution:-

$$\mathcal{L}(\cosh at) = \frac{1}{2}\left(\mathcal{L}(e^{at}) + \mathcal{L}(e^{-at})\right)$$

$$= \frac{1}{2}\left(\frac{1}{s-a} + \frac{1}{s+a}\right)$$

$$= \frac{s}{s^2-a^2}$$

$$\mathcal{L}(\sinh at) = \frac{1}{2}\left(\mathcal{L}(e^{at}) - \mathcal{L}(e^{-at})\right)$$

$$= \frac{1}{2}\left(\frac{1}{s-a} - \frac{1}{s+a}\right)$$

$$= \frac{a}{s^2-a^2}$$

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Ex:- "6.3"

Q Write the following using unit step functions and find its transform

$$f(t) = \begin{cases} 2 & \text{if } 0<t<1 \\ \frac{1}{2}t^2 & \text{if } 1<t<\frac{1}{2}\pi \\ \cos t & \text{if } t>\frac{1}{2}\pi \end{cases}$$

• Solution:-

• Step "1":-

In terms of unit step function.

$$f(t) = 2(1-u(t-1)) + \frac{1}{2}t^2\left(u(t-1) - u\left(t-\frac{1}{2}\pi\right)\right)$$

$$+ (\cos t)\,u\left(t-\frac{1}{2}\pi\right).$$

• Step "2":-

$$\mathcal{L}\left\{\frac{1}{2}t^2 u(t-1)\right\}$$

$$= \mathcal{L}\left\{\left(\frac{1}{2}(t-1)^2 + (t-1) + \frac{1}{2}\right)u(t-1)\right\}$$

$$= \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\right)e^{-s}$$

$$\mathcal{L}\left\{\frac{1}{2}t^2 u\left(t-\frac{1}{2}\pi\right)\right\}$$

$$\mathcal{L} = \left\{\frac{1}{2}\left(t-\frac{1}{2}\pi\right)^2 + \frac{\pi}{2}\left(t-\frac{1}{2}\pi\right) + \frac{\pi^2}{8}\right) u\left(t-\frac{1}{2}\pi\right)\right\}$$

$$\mathcal{L} = \left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s}\right) e^{-\pi s/2}$$

$$\mathcal{L}\left\{(\cos t)\, u\left(t-\frac{1}{2}\pi\right)\right\}$$

$$= \mathcal{L}\left\{-\left(\sin\left(t-\frac{1}{2}\pi\right)\right) u\left(t-\frac{1}{2}\pi\right)\right\}$$

$$= -\frac{1}{s^2+1} e^{-\pi s/2}$$

Together,

$$\mathcal{L}(f) = \frac{2}{s} - \frac{2}{s}e^{-s} + \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\right)e^{-s} - \left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s}\right)$$

$$e^{-\pi s/2} - \frac{1}{s^2+1} e^{-\pi s/2}$$ Ans.

* Example "2" :-

Q Find the inverse transform $f(t)$ of

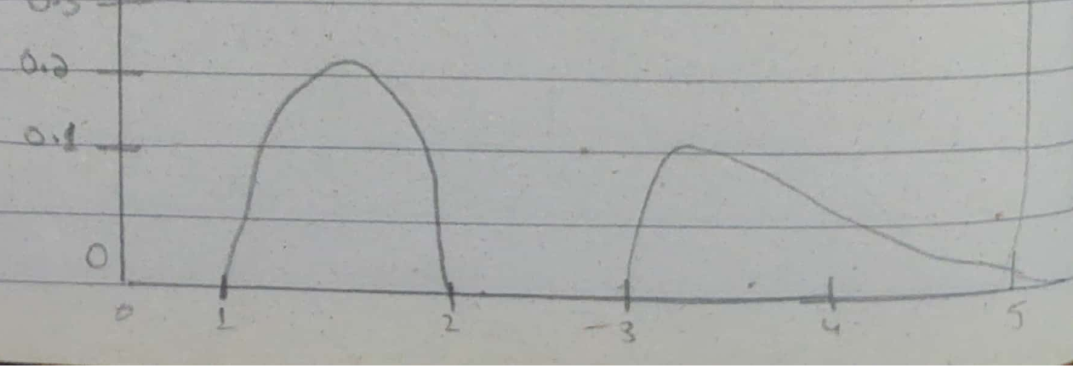
$$F(s) = \frac{e^{-s}}{s^2+\pi^2} + \frac{e^{-2s}}{s^2+\pi^2} + \frac{e^{-3s}}{(s+2)^2}$$

• Solution:-

Without the exponential functions in the numerator the three terms of $F(s)$ would have the inverses $(\sin \pi t)/\pi$, $(\sin \pi t)/\pi$, and te^{-2t} because $1/s^2$ has the inverse t, so that $1/(s+2)^2$ has the inverse te^{-2t} by the first shifting theorem in Sec.

$$f(t) = \frac{1}{\pi} \sin(\pi(t-1))\, u(t-1) + \frac{1}{\pi} \sin(\pi(t-2))\, u(t-2) + (t-3)\, e^{-2(t-3)}\, u(t-3).$$

Now $\sin(\pi t - \pi) = -\sin \pi t$ and $\sin(\pi t - 2\pi) = \sin \pi t$



Ex:- "11.9"

* Fourier Transform:-

* Example "1":-

Q Find the fourier tranform of $f(x) = 1$ if $|x| < 1$ and $f(x) = 0$ otherwise

• Solution:-

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-i\omega x} \, dx = \frac{1}{\sqrt{2\pi}} \cdot \frac{e^{-i\omega x}}{-i\omega} \Big|_{-1}^{1}$$

$$= \frac{1}{-i\omega\sqrt{2\pi}} \left(e^{-i\omega} - e^{i\omega}\right).$$

As in (3) we have $e^{i\omega} = \cos\omega + i\sin\omega$, $e^{-i\omega} = \cos\omega - i\sin\omega$, and by subtraction

$$e^{i\omega} - e^{-i\omega} = 2i\sin\omega$$

After Substitution,

$$\hat{f}(\omega) = \sqrt{\frac{\pi}{2}} \, \frac{\sin\omega}{\omega}.$$

* **Example "3":-**

Q Find the fourier transform of xe^{-x^2} from table.

Solution:-

$$F(xe^{-x^2}) = F\left\{-\frac{1}{2}\left(e^{-x^2}\right)'\right\}$$

$$= -\frac{1}{2} F\left\{\left(e^{-x^2}\right)'\right\}$$

$$= -\frac{1}{2}\, i\omega F\left(e^{-x^2}\right)$$

$$= -\frac{1}{2}\, i\omega \frac{1}{\sqrt{2}} e^{-\omega^2/4}$$

$$= -\frac{i\omega}{2\sqrt{2}} e^{-\omega^2/4}$$

Ans.

* FOURIER SERIES:-

Q) $f(x) = \begin{cases} 0, & -\pi < x < -\pi/2 \\ 1, & -\pi/2 < x < \pi/2 \\ 0, & \pi/2 < x < \pi \end{cases}$

• Solution:-

let,

$\because f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$

$\because a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx \quad , \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x).\cos nx\,dx$

$\because b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x).\sin nx\,dx$

$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx$

$= \frac{1}{\pi} \left[\int_{-\pi}^{-\pi/2} 0\,dx + \int_{-\pi/2}^{\pi/2} 1\,dx \; \int_{\pi/2}^{\pi} 0\,dx \right]$

$= \frac{1}{\pi} \left[0 + \int_{-\pi/2}^{\pi/2} dx + 0 \right]$

$$a_0 = \frac{1}{\pi}\left[x\right]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{\pi}\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi}{2} + \frac{\pi}{2}\right]$$

$$= \frac{2\pi}{2\pi}$$

$$\boxed{a_0 = 1}$$

$$\because \; a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot \cos nx \, dx.$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} 0\cdot \cos nx\,dx + \int_{-\pi/2}^{\pi/2} 1\cdot \cos nx\,dx + \int_{\pi/2}^{\pi} 0\cdot \cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[0 + \int_{-\pi/2}^{\pi/2} \cos nx\,dx + 0\right]$$

$$a_n = \frac{1}{\pi}\left[\int_{-\pi/2}^{\pi/2} \cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[\frac{\sin nx}{n}\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \frac{1}{\pi n}\Big[\sin nx\Big]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{n\pi}\left[\sin n\left(\frac{\pi}{2}\right) - \sin n\left(-\frac{\pi}{2}\right)\right]$$

$$= \frac{1}{n\pi}\left[\sin n\frac{\pi}{2} + \sin n\frac{\pi}{2}\right]$$

$$= \frac{1}{n\pi}\, 2\sin n\frac{\pi}{2}$$

$\frac{\pi}{2} = 90$

$\sin 90 = 1$

$$\boxed{a_n = \frac{2}{n\pi}}$$

$$\because b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot \sin nx\, dx$$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} 0\,\sin nx\, dx + \int_{-\pi/2}^{\pi/2} 1\cdot \sin nx\, dx + \int_{\pi/2}^{\pi} 0\cdot \sin nx\, dx\right]$$

$$b_n = \frac{1}{\pi}\left[0 + \int_{-\pi/2}^{\pi/2} \sin nx\, dx + 0\right]$$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi/2}^{\pi/2} \sin nx\, dx\right]$$

$$b_n = \frac{1}{\pi}\left[-\frac{\cos nx}{n}\right]_{-\pi/2}^{\pi/2}$$

$\cancel{\cos \frac{n\pi}{2} - \cos \frac{n\pi}{2}}$

$$b_n = \frac{-1}{n\pi}\left[\cos n\left(\frac{\pi}{2}\right) - \cos n\left(\frac{-\pi}{2}\right)\right]$$

$\because \cos(-\theta)$
$= \cos\theta$

$$\boxed{b_n = 0}$$

$$\because f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right)$$

$$f(x) = \frac{1}{2}(1) + \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} \cos nx + 0\right)$$

$$= \frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} \cos nx\right)$$

$$= \frac{1}{2} + \frac{2}{1\pi} \cos x + \frac{2}{2\pi} \cos 2x + \frac{2}{3\pi} \cos 3x + \frac{2}{4\pi} \cos 4x$$

$$+ \frac{2}{5\pi} \cos 5x + \cdots\cdots$$

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \cos x + \frac{1}{\pi} \cos 2x + \frac{2}{3\pi} \cos 3$$

$$+ \frac{1}{2\pi} \cos 4x + \frac{2}{5\pi} \cos 5x + \cdots\cdots$$

Ans.

$a_0 = 2\pi, \; a_n = 0$
$b_n = -\frac{2}{n}$

Q $f(x) = x \;, \quad 0 < x < 2\pi$

• Solution:-

$$\because \; a_0 = \frac{1}{\pi} \int_{\pi}^{-\pi} f(x)\,dx$$

$$a_0 = \frac{1}{\pi} \int_{0}^{2\pi} x \, dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{2\pi}$$

$$a_0 = \frac{1}{2\pi} \left[\frac{(2\pi)^2}{\cancel{2}} - (0)^2 \right]$$

$$a_0 = \frac{1}{2\pi} \left[4\pi^2 - 0 \right]$$

$$a_0 = \frac{1}{2\pi} \left[4\pi^2 \right]$$

$$a_0 = \frac{4\pi^2}{2\pi} \Rightarrow \boxed{a_0 = 2\pi}$$

$$\because a_n = \frac{1}{\pi}\int_{\pi}^{-\pi} f(x).\,\text{Cos}\,nx\,dx$$

$$a_n = \frac{1}{\pi}\int_{0}^{2\pi} \underbrace{\underline{x.\ \text{Cos}\,nx\,dx}}_{U.V}$$

$$\because \int U.V = U.\ \text{Integral of } V - \int (\text{derivative of } U)(\text{Integral of } V)\,dx$$

$$a_n = \frac{1}{\pi}\left[(x)\left(\frac{\text{Sin}\,nx}{n}\right) - \int_0 (1)\left(\frac{\text{Sin}\,nx}{n}\right) dx\right]$$

$$a_n = \frac{1}{n\pi}\left[x\,\text{Sin}\,nx + \frac{1}{n}.\frac{\text{Cos}\,nx}{n}\right]_0^{2\pi}$$

$$a_n = \frac{1}{n\pi}\left[x\,\text{Sin}\,nx + \frac{1}{n^2}.\ \text{Cos}\,nx\right]_0^{2\pi}$$

$$a_n = \frac{1}{n\pi}\left[(2\pi)\text{Sin}\,n(2\pi) - (0)\text{Sin}\,n(0)\right] +$$

$$\frac{1}{n^2}\left[\text{Cos}\,n(2\pi) - \text{Cos}\,n(0)\right]$$

$$a_n = \frac{1}{n\pi}\left[0 - 0\right] + \frac{1}{n^2}\left[1 - 1\right]$$

$$\boxed{a_n = 0}$$

$$\because \; b_n = \frac{1}{\pi} \int_{\pi}^{\pi} f(x) \cdot \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \underset{U}{x} \cdot \underset{V}{\sin nx} \, dx$$

$$\because \int U.V = U \cdot (\text{Integral of } V) - \int (\text{derivative of } U)(\text{Integral of } V) \, dx$$

$$b_n = \frac{1}{\pi} \left[(x) \left(-\frac{\cos nx}{n} \right) - \int_0 (1) \left(-\frac{\cos nx}{n} \right) dx \right]$$

$$b_n = -\frac{1}{n\pi} \left[x \cos nx + \frac{1}{n} \cdot \frac{\sin nx}{n} \right]_0^{2\pi}$$

$$b_n = -\frac{1}{n\pi} \left[x \cos nx + \frac{1}{n^2} \cdot \sin nx \right]_0^{2\pi}$$

$$b_n = -\frac{1}{n\pi} \left[(2\pi) \cos n(2\pi) - (0) \cos n(0) \right] + \frac{1}{n^2}$$

$$\left[\sin n(2\pi) - \sin n(0) \right]$$

$$b_n = -\frac{1}{n\pi} \left[2\pi(1) - 0 \right] + \frac{1}{n^2} \left[0 - 0 \right]$$

$$b_n = \frac{-1}{n\pi}\left[2\pi\right]$$

$$b_n = \frac{-2\cancel{\pi}}{n\cancel{\pi}}$$

$$\boxed{b_n = \frac{-2}{n}}$$

Now,

Putting values of a_0, a_n & b_n in equation below

$$\because f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left(a_n \cos nx + b_n \sin nx\right)$$

$$f(x) = \frac{1}{2}(2\pi) + \sum_{n=1}^{\infty}\left(0 \cos nx + \left(\frac{-2}{n}\right)\sin nx\right)$$

$$f(x) = \frac{\cancel{2}\pi}{\cancel{2}} + \sum_{n=1}^{\infty}\left(0 - \frac{2}{n}\sin nx\right)$$

$$f(x) = \pi - \sum_{n=1}^{\infty}\left(\frac{2}{n}\sin nx\right)$$

$$f(u) = \frac{\pi - 2}{1} \sin x + \frac{\pi - 2}{2} \sin 2x$$

$$+ \frac{\pi - 2}{3} \sin 3x + \frac{\pi - 2}{4} \sin 4x$$

$$+ \frac{\pi - 2}{5} \sin 5x + \dots$$

$$f(u) = \pi - 2 \sin x + \pi - 1 \sin 2u + \frac{\pi - 2}{3} \sin 3x$$

$$+ \pi - \frac{1}{2} \sin 4u + \frac{\pi - 2}{5} \sin 5u + \dots$$

Ans.

$$f(u) = \pi - 2\sin x - \sin 2x - \frac{2}{3} \sin 3x$$

$$- \frac{\cancel{\pi}}{2} \sin 4x - \frac{2}{5} \sin 5x + \dots$$